Pranav Rathod

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Education

University of Southern California — January 2024 – December 2025
Master of Science in Computer Science, Multimedia and Creative Technologies — *Los Angeles, CA*

University of Illinois at Chicago — August 2019 – May 2023
Bachelor of Science in Computer Science, Minor in Art — *Chicago, IL*

Relevant Coursework

Analysis of Algorithms, Multimedia Systems Design, Computer Graphics, Computer Animation and Simulation, Game Engine Development, 3D Graphics and Rendering, Augmented/Virtual Reality, Artificial Intelligence, Framework-based Development

Skills

Languages: C/C++, Python, C#, Java, JavaScript, SQL, Dart, Go, F#
Technical: OOP, Git, CI/CD (GitHub Actions), Docker, Terraform, Linux/Unix, OpenGL, PyTorch, Node.js, React, HTML5, CSS3, Unity, WebGL, Firebase, GCP, SQLite3, Multi-Threading, TCP/IP, Testing Frameworks (Google/JUnit5)

Experience

Media Support Specialist — January 2024 – Present
Annenberg TechOps Information Technology, University of Southern California — *Los Angeles, CA*
- Troubleshoot cross-platform rendering and asset pipeline issues, ensuring reliable deployment of multimedia applications
- Resolve web and multimedia workflow issues by troubleshooting HTML, WordPress, and Adobe Creative Cloud
- Authored technical documentation and tutorials to streamline multimedia production for 200+ students and faculty

Research Assistant — June 2022 – December 2023
ELiCIT Lab, University of Illinois at Chicago — *Chicago, IL*
- Collaborated with Adobe Research to prototype a Flutter-based mobile app for multimodal interaction studies
- Explored backend trade-offs in storing synchronized multimodal data for use in training NUI models
- Developed UI workflows and backend logic to manage speech and gesture-driven image editing tasks across 12 workflows
- Integrated GCP Speech-to-Text API and installed real-time gesture logging using touch coordinates and timing data

Teaching Assistant — January 2022 – December 2022
Department of Computer Science, University of Illinois at Chicago — *Chicago, IL*
- Led community discussions and labs covering recursion, graph traversal, and dynamic programming for 100+ students
- Guided students in debugging and writing code in C, C++ and building applications using Flutter, Dart across 3 semesters
- Formulated course logistics and graded assignments for intermediate and advanced Computer Science courses each semester

Projects

PrimeEngine | *C++, OpenGL, Maya* — April 2025
- Built a scalable real-time 2D overlay and player controls to enhance debugging efficiency in a custom game engine
- Optimized rendering performance by 40% via algorithmic AABB frustum culling and integrated real-time visualization tools
- Integrated Maya asset workflow and optimized render loop to support faster design-to-engine iteration cycles

Motion Capture Interpolation | *C++, OpenGL, FLTK* — March 2025
- Reconstructed missing motion capture sequences using Linear, Bezier, SLERP, and Bezier SLERP interpolation methods
- Evaluated performance versus accuracy trade-offs in interpolation methods under real-time compute and memory constraints
- Implemented OOP-based playback system with FLTK UI to compare interpolation quality in real time

Segmented Video Compression | *Python, Detectron2, OpenCV, PyTorch* — December 2024
- Implemented block-based encoder using DCT, quantization, and motion estimation to compress .rgb videos efficiently
- Achieved 95% accuracy in foreground–background separation with Detectron2 and optical flow for adaptive compression
- Built synchronized A/V player supporting 30 FPS playback with real-time IDCT decoding and pause/step controls

Traffic Crash Analysis Using ML | *Python, Pandas, JavaScript* — Apr 2023
- Trained Naive Bayes classifier on 1M+ records using weather, lighting, and road surface conditions to predict crash types
- Visualized geospatial heat maps and frequency analyses showing seasonal crash trends and high-risk areas in Chicago
- Examined crash data by vehicle type and demographics, identifying higher risk in sport vehicles and drivers aged 25–30

Achievements & Leadership

SIGGRAPH Student Volunteer Team Leader | *Denver, CO; Tokyo, Japan; Vancouver, BC* — 2024–2025
- Oversaw operations for 350+ international student volunteers at world's premier computer graphics conference
- Facilitated event logistics and attendee support, collaborating with global teams of students and industry professionals

Undergraduate Research Forum | *Chicago, IL* — April 2023
- Achieved 3rd place in the Engineering and Physical Sciences category for presenting NUI-based research

Engineering Expo | *Chicago, IL* — April 2021
- Won 'Best in Category' for designing an Arduino-based device to enforce COVID-19 social distancing